

SOUTHEAST HAWAII BUOY 51004, USA

WMO: 992620

Lat: 17.533N

Lon: 152.255W

Elev: 0

StdP: 14.70

Time Zone: -10.00 (W10)

Period: 94-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	70.8	71.7	N/A	N/A	N/A	N/A	N/A	N/A	30.0	73.5	26.7	73.9	18.2	60	0.700

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	2.8	83.6	N/A	82.9	N/A	82.0	N/A	N/A	N/A	N/A	N/A	N/A	N/A	11.4	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 26.2	24.6	23.4	DB	67.0	81.8	4.7	2.1	63.6	83.4	60.8	84.6	58.2	85.8	54.8	87.3	(4)
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	76.5	75.1	74.6	74.7	75.2	76.1	76.9	77.7	78.6	78.5	78.0	77.1	75.7
	DBStd	2.10	1.39	1.61	1.51	1.51	1.64	1.96	1.89	1.58	1.52	1.37	1.33	1.24
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	9676	777	688	765	755	809	808	859	885	854	868	812	796
	CDD65	4201	312	268	300	305	344	358	394	420	404	403	362	331
	CDH74	25703	1101	812	926	1209	1823	2400	3051	3625	3444	3248	2477	1587
	CDH80	797	0	0	0	0	15	142	174	214	151	76	24	1
Wind	WSAvg	16.9	16.4	17.0	16.8	17.8	16.0	16.5	16.9	17.2	15.6	16.7	17.7	18.5
Precipitation	PrecAvg	14.2	1.9	1.5	1.8	0.5	0.4	0.2	0.6	1.1	1.2	1.3	1.8	1.9
	PrecMax	29.6	4.7	6.3	7.5	3.1	2.5	1.2	2.1	8.0	5.4	3.8	5.6	11.7
	PrecMin	6.1	0.0	0.2	0.1	0.0	0.0	0.1	0.1	0.1	0.1	0.1	0.0	0.0
	PrecStd	5.6	1.5	1.3	1.8	0.6	0.5	0.2	0.4	1.5	1.4	0.8	1.4	2.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	79.5	79.0	79.2	79.4	81.4	84.3	84.2	84.1	83.9	82.8	82.4	80.1
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	78.4	78.1	78.2	78.5	80.4	83.5	83.3	83.2	82.4	81.9	80.5	78.9
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	77.9	77.4	77.5	78.2	79.7	82.2	82.4	82.4	81.8	80.9	79.7	78.3
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	77.3	77.2	77.2	77.7	78.9	79.5	80.6	81.3	80.8	80.3	79.2	77.9
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	3.1	3.0	2.9	3.0	3.0	3.2	3.1	2.8	2.7	2.6	2.5	2.8
		MCDBR	4.0	3.2	3.3	4.1	3.7	3.7	3.5	3.9	4.2	3.7	3.4	3.6
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub	0.359	0.365	0.372	0.404	0.396	0.382	0.382	0.384	0.373	0.372	0.372	0.362	
	taud	2.606	2.588	2.569	2.443	2.486	2.519	2.543	2.546	2.566	2.609	2.605	2.621	
	Ebn at Noon	295	298	298	286	285	286	286	288	293	292	289	291	
	Edh at Noon	29	31	33	37	35	34	33	33	32	30	29	28	
All-Sky Solar Radiation	RadAvg	1517	1676	1881	2079	2173	2165	2183	2151	2070	1766	1482	1404	
	RadStd	84	106	158	127	89	93	72	79	82	68	107	104	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page